# Opera ng System Lab Observa on

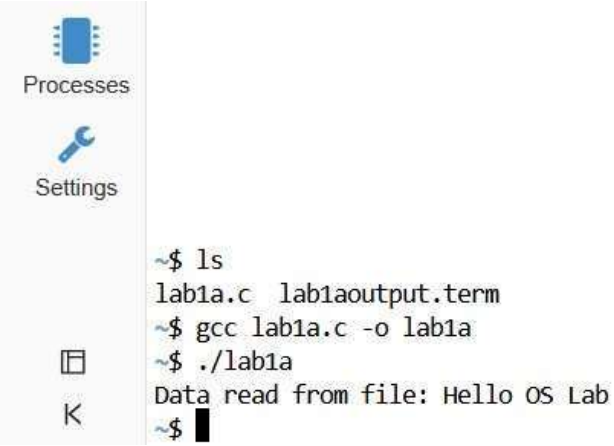
1a

Program:

```
#include <stdio.h>

#include <unistd.h>

#include <fcntl.h>


int main()

{     int fd;

char buffer[50];


    /* open() – create and open a file */     fd =

open("sample.txt", O_CREAT | O_RDWR, 0644);


    if (fd < 0)

    {

        prin ("File opening failed\n");

return 1;

    }


    /* write() – write data into file */

write(fd, "Hello OS Lab", 12);


    /* move file pointer to beginning */

lseek(fd, 0, SEEK_SET);
```

```
    /* read() – read data from file */
read(fd, buffer, 12);

    /* display data */     prin ("Data read from
file: %s\n", buffer);

    /* close() – close the file */
close(fd);

    return 0;
}
```

Output:



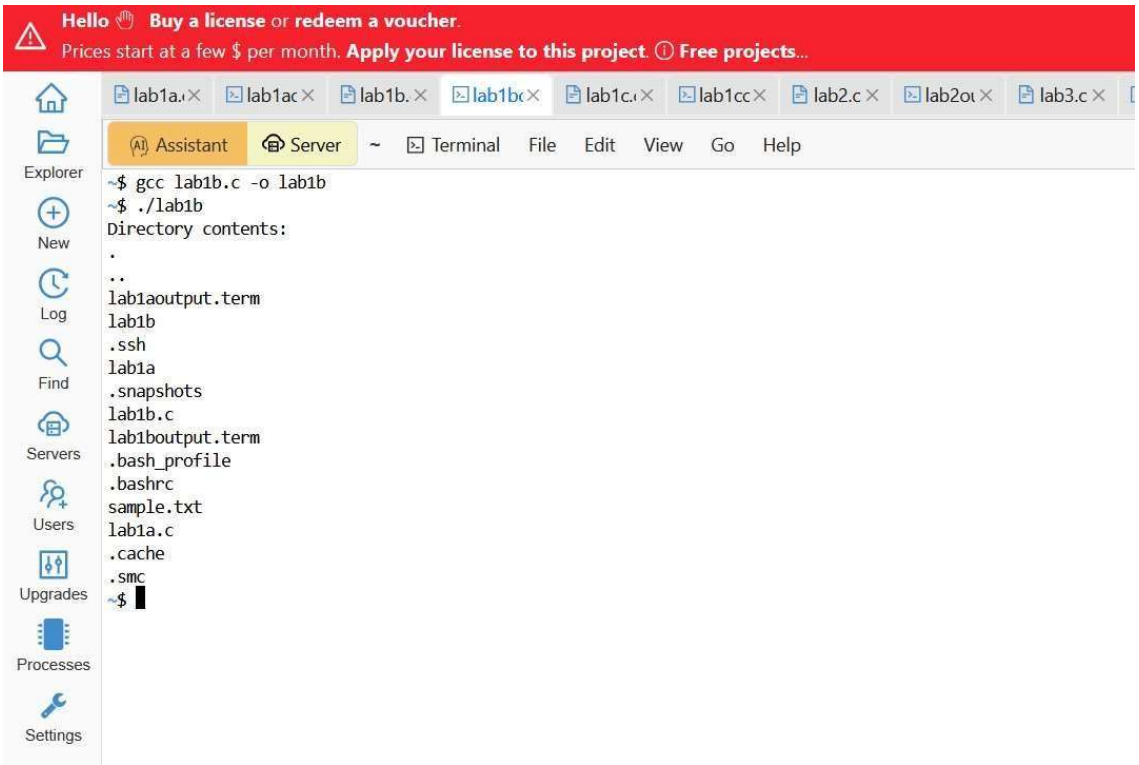

1b:

Program:

```
#include <stdio.h>
#include <dirent.h>

int main()
```

```
{
   DIR *d;     struct
dirent *entry;

   /* open current directory */     d
= opendir(".");

   if (d == NULL)
   {
      prin ("Directory cannot be opened\n");
      return 1;
   }
   /* read directory entries */     prin
("Directory contents:\n");     while
((entry = readdir(d)) != NULL)
   {
      prin ("%s\n", entry->d_name);
   }
   /* close directory */
closedir(d);     return
0;
}
```

Output:



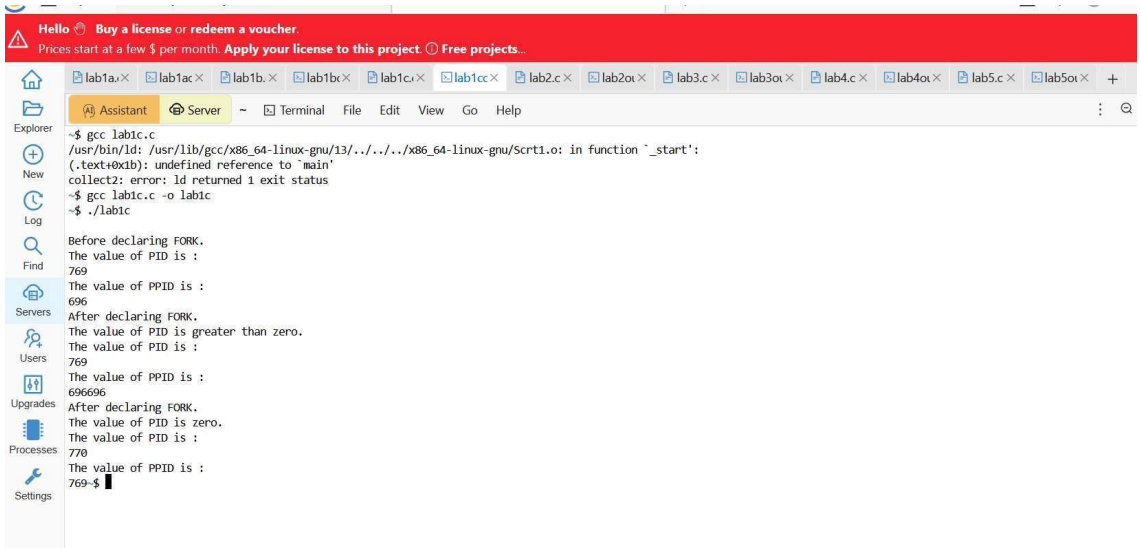

1c:

Program:

```
#include<stdio.h>

#include<unistd.h>

#include<sys/types.h> void

main()

{

prin ("\nBefore declaring FORK.");

prin ("\nThe value of PID is : "); prin

("\n%d",getpid()); prin ("\nThe

value of PPID is : ");

prin ("\n%d",getppid());


pid_t pid=fork();//Used to create child process if(pid==0)//Child

Process
```

```
{
prin ("\nA er declaring FORK."); prin
("\nThe value of PID is zero."); prin
("\nThe value of PID is : "); prin
("\n%d",getpid()); prin ("\nThe
value of PPID is : ");
prin ("\n%d",getppid());
}
else if(pid>0)//Parent Process
{
prin ("\nA er declaring FORK."); prin ("\nThe
value of PID is greater than zero."); prin ("\nThe
value of PID is : "); prin ("\n%d",getpid()); prin
("\nThe value of PPID is : "); prin
("\n%d",getppid());
}
else//Error
{
prin ("\nA er declaring FORK."); prin ("\nThe
value of PID is less than zero."); prin ("\nThe
value of PID is : "); prin ("\n%d",getpid());
prin ("\nThe value of
PPID is : "); prin ("%d",getppid());
}

}
```

Output:




Lab2:

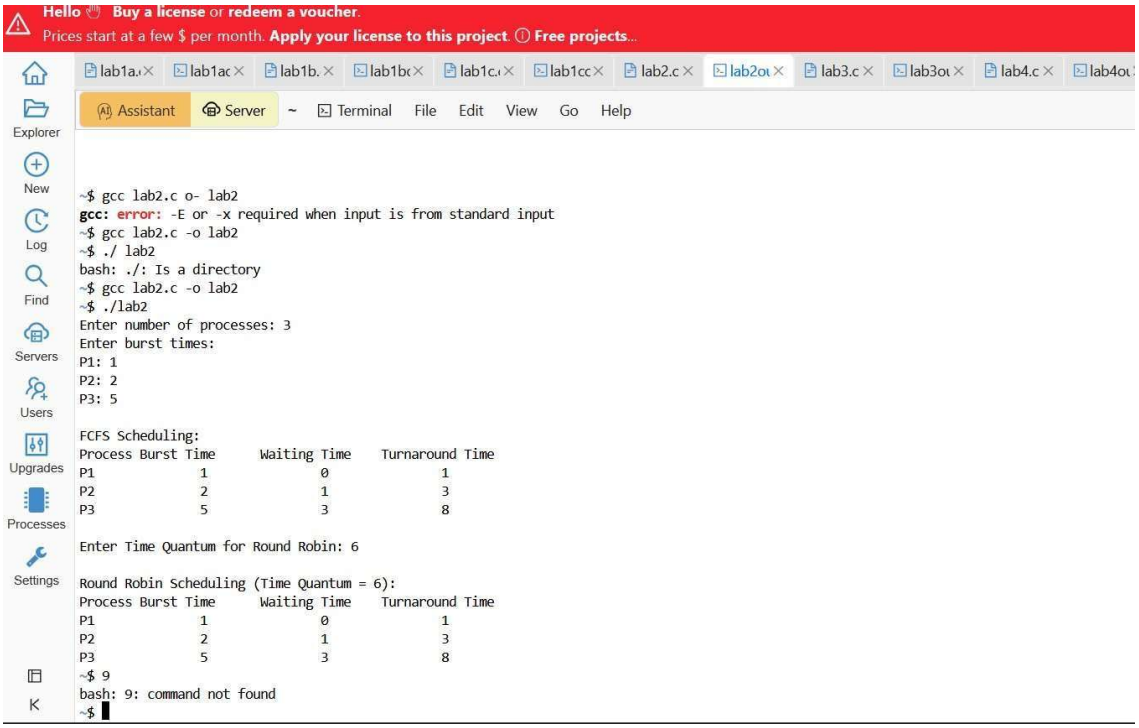
Program:

```
#include <stdio.h>  // FCFS Scheduling   void fcfs(int n, int bt[]) {
int wt[10], tat[10];  wt[0] = 0;  for(int i = 1; i < n; i++)  wt[i] = wt[i-
1] + bt[i-1];  prin ("\nFCFS Scheduling:\n");  prin ("Process\tBurst
Time\tWai ng Time\tTurnaround Time\n");
for(int i = 0; i < n; i++) {   tat[i] = wt[i] + bt[i];  prin

("P%d\t\t%d\t\t%d\t\t%d\n",

i+1, bt[i], wt[i], tat[i]);
}
}
// Round Robin Scheduling  void
roundRobin(int n, int bt[], int tq) {  int
rem_bt[10], wt[10], tat[10];  int t = 0;
```

```
for(int i = 0; i < n; i++) {  rem_bt[i] = bt[i];
wt[i] = 0;
}
while(1) {  int done = 1;
for(int i = 0; i < n; i++) {
if(rem_bt[i] > 0) {  done =
0;  if(rem_bt[i] > tq) {  t
+= tq;
rem_bt[i] -= tq;
                } else {
t += rem_bt[i];
wt[i] = t - bt[i];
rem_bt[i] = 0;
                }
            }
        }
        if(done == 1)
break;
    }

    prin ("\nRound Robin Scheduling (Time Quantum = %d):\n", tq);      prin ("Process\tBurst
Time\tWai ng Time\tTurnaround Time\n");

    for(int i = 0; i < n; i++) {          tat[i] = bt[i] + wt[i];
prin ("P%d\t\t%d\t\t%d\t\t%d\n", i+1, bt[i], wt[i], tat[i]);
    }
}
```

```
int main() {
int n, tq;      int
bt[10];

    prin ("Enter number of processes: ");
scanf("%d", &n);  prin ("Enter burst
  mes:\n");  for(int i = 0; i < n; i++) {
prin ("P%d: ", i+1);  scanf("%d",
&bt[i]);
}
fcfs(n, bt);  prin ("\nEnter Time Quantum for Round
Robin: ");  scanf("%d", &tq);  roundRobin(n, bt, tq);
return 0;

}
```

Output:



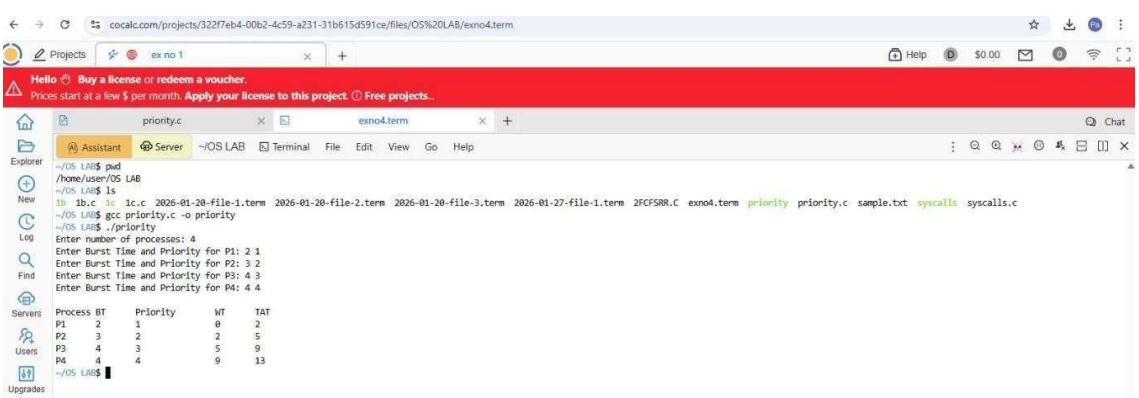

Lab3:

Program:

```
#include <stdio.h>

int main() { int
n, i, j;
int bt[10], pr[10], p[10]; int
wt[10], tat[10], temp;

prin ("Enter number of processes: "); scanf("%d",
&n);

for(i = 0; i < n; i++) {
p[i] = i + 1; prin \
```

```
 Enter Burst Time and Priority for P%d: ", p[i]); scanf("%d
%d", &bt[i], &pr[i]);
}
// Sort by Priority for(i = 0; i < n; i++) {
for(j = i + 1; j < n; j++) { if(pr[i] > pr[j])
{ temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
temp = p[i]; p[i] = p[j]; p[j] = temp;
}
}
}
wt[0] = 0; for(i = 1; i < n; i++) wt[i] = wt[i - 1] + bt[i - 1]; for(i = 0;
i
< n; i++) tat[i] = wt[i] + bt[i]; prin
("\nProcess\tBT\tPriority\tWT\tTAT\n"); for(i = 0; i < n; i++) prin
("P%d\t%d\t%d\t\t%d\t%d\n", p[i], bt[i], pr[i], wt[i], tat[i]); return
0;

}
```

Output:



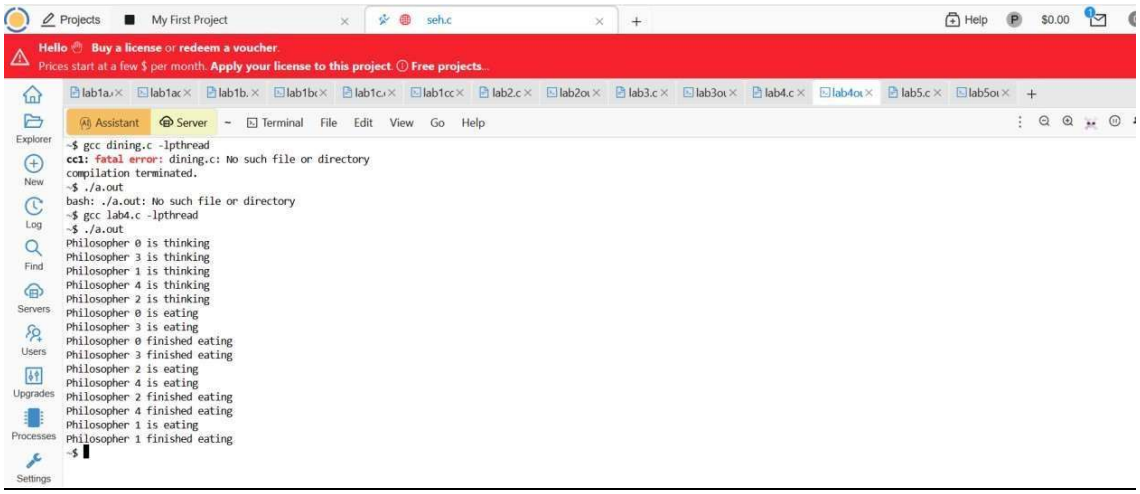
Lab4:

Program:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h> #define N 5 sem_t
forks[N]; pthread_t philosophers[N]; void*
philosopher(void* num) { int id = *(int*)num;
prin ("Philosopher %d is thinking\n", id);
sleep(1); sem_wait(&forks[id]);
sem_wait(&forks[(id + 1) % N]); prin
("Philosopher %d is ea ng\n", id); sleep(1);
sem_post(&forks[id]); sem_post(&forks[(id
+ 1) % N]); prin ("Philosopher %d finished ea
ng\n",
id); return NULL;
}
int main() {
int i, id[N];

for (i = 0; i < N; i++)
sem_init(&forks[i], 0, 1);
for (i = 0; i < N; i++) {
id[i] = i;
pthread_create(&philosophers[i], NULL, philosopher, &id[i]);
```

```
    }    for (i = 0; i <
N; i++)
pthread_join(philosophers[i], NULL);    return 0;
}
```

Output:




Lab5:

Program:

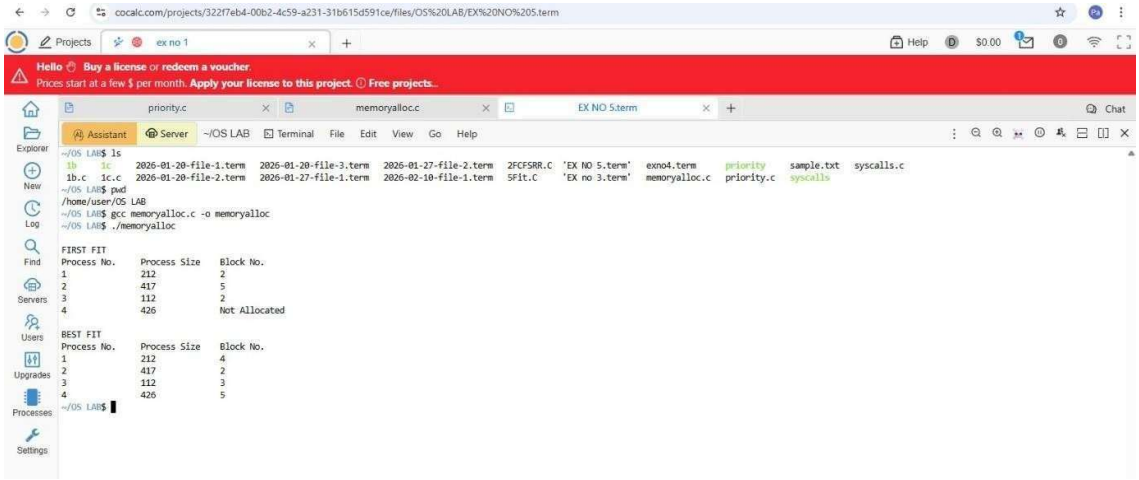
```
#include<stdio.h>  void firstFit(int blockSize[], int m, int
processSize[], int n)
{
int alloca on[n];
for(int i=0;i<n;i++)
alloca on[i]=-1;

    for(int i=0;i<n;i++)
    {
        for(int j=0;j<m;j++)
```

```
    {
        if(blockSize[j] >= processSize[i])
        {   alloca
on[i] = j;
blockSize[j] -=
processSize[i];
break;

        }
    }
  }

  prin ("\nFIRST FIT\nProcess No.\tProcess Size\tBlock No.\n");
for(int i=0;i<n;i++)
  {
    if(alloca on[i]!=-1)  prin
("%d\t\t%d\t\t%d\n",i+1,processSize[i],alloca on[i]+1);  else
prin ("%d\t\t%d\t\tNot Allocated\n",i+1,processSize[i]);
  }
}

void bestFit(int blockSize[], int m, int processSize[], int n)
{
  int alloca on[n];
for(int i=0;i<n;i++)
alloca on[i]=-1;

  for(int i=0;i<n;i++)
```

```
    {
        int bestIdx=-1;
for(int j=0;j<m;j++)
        {
            if(blockSize[j]>=processSize[i])
            {
                if(bestIdx==-1 || blockSize[j]<blockSize[bestIdx])
bestIdx=j;
            }
        }

        if(bestIdx!=-1)
        {
            alloca on[i]=bestIdx;   blockSize[bestIdx]-
=processSize[i];
        }
    }

    prin ("\nBEST FIT\nProcess No.\tProcess Size\tBlock No.\n");
for(int i=0;i<n;i++)
    {
        if(alloca on[i]!=-1)   prin
("%d\t\t%d\t\t%d\n",i+1,processSize[i],alloca on[i]+1);   else
prin ("%d\t\t%d\t\tNot Allocated\n",i+1,processSize[i]);
    }
}

int main()
```

```
{
    int blockSize[] = {100, 500, 200, 300, 600};
int processSize[] = {212, 417, 112, 426};

    int m = 5;
int n = 4;    int
block1[5],
block2[5];
for(int
i=0;i<m;i++)
    {
        block1[i]=blockSize[i];
block2[i]=blockSize[i];
    }

    firstFit(block1,m,processSize,n);
bestFit(block2,m,processSize,n);

    return 0;
}
```

Output:



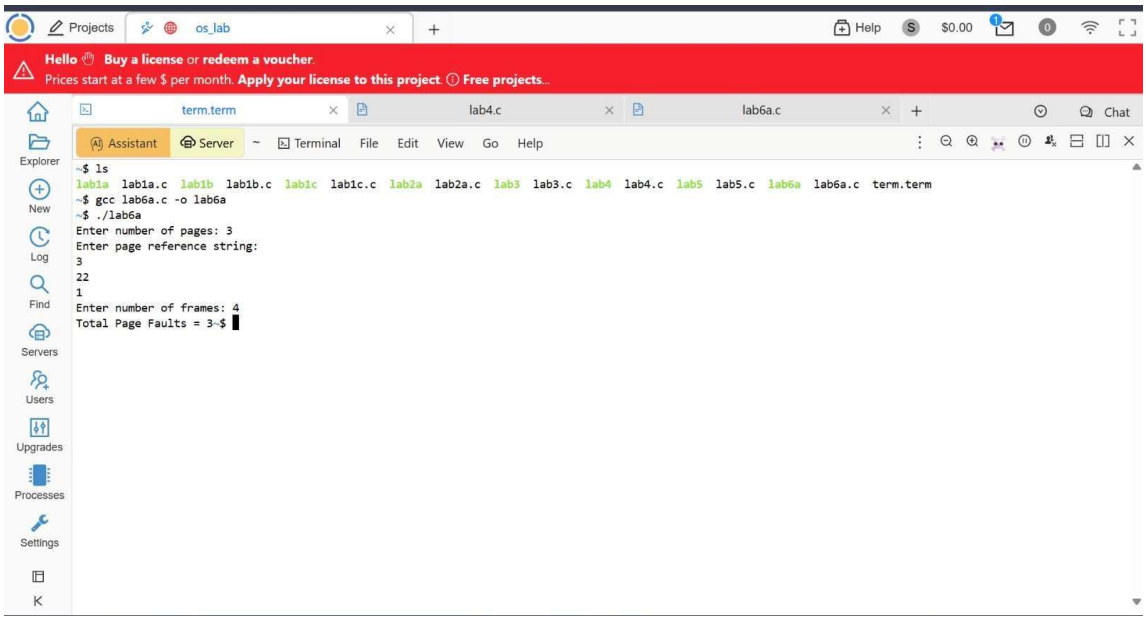

LAB 6A

PROGRAM:-

```
#include <stdio.h>
int main()
{
   int pages[50], frames[10], me[10];       int n, f, i, j,
k, flag, faults = 0, pos, min, counter = 0;

   prin ("Enter number of pages: ");
scanf("%d",&n);

   prin ("Enter page reference string:\n");
for(i=0;i<n;i++)
scanf("%d",&pages[i]);
```

```
   prin ("Enter number of frames: ");
scanf("%d",&f);

   for(i=0;i<f;i++)
frames[i] = -1;

   for(i=0;i<n;i++)
   {
      flag = 0;   // <-- FIXED here
for(j=0;j<f;j++)
      {
         if(frames[j]==pages[i])
         {
            counter++;
me[j]=counter;                  flag = 1;
// <-- FIXED here                break;
         }
      }
      if(flag==0)
      {
if(i<f)
         {
            frames[i]=pages[i];
counter++;
me[i]=counter;
faults++;
         }
else
         {
```

```
            min= me[0];
pos=0;
for(k=1;k<f;k++)
            {
               if( me[k]<min)
               {
                  min= me[k];
                  pos=k;
               }
            }
            frames[pos]=pages[i];
counter++;
me[pos]=counter;
            faults++;
         }
      }
   }
   prin ("Total Page Faults = %d",faults);
return 0;
}
```




LAB 6B

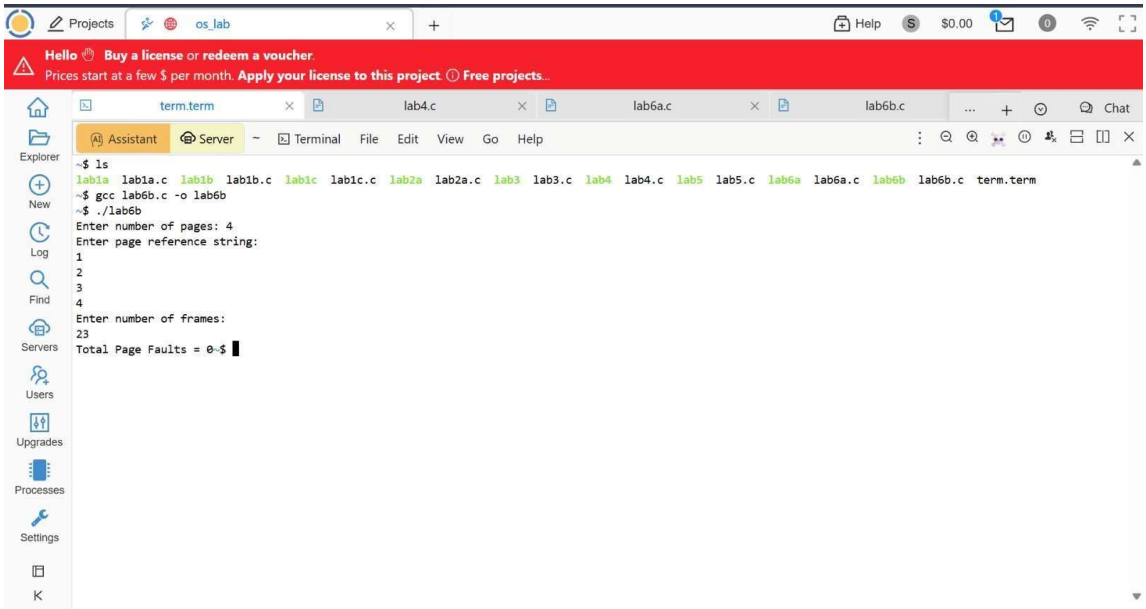
PROGRAM:-

```
#include <stdio.h>

int main() {     int pages[50],
frames[10];
   int n, f, i, j, k, flag, faults = 0;
int pos, farthest, index;

   // Input: number of pages     prin
("Enter number of pages: ");
scanf("%d", &n);

   // Input: page reference string     prin
("Enter page reference string:\n");
```

```
    for (i = 0; i < n; i++)
scanf("%d", &pages[i]);

    // Input: number of frames     prin
("Enter number of frames: ");
scanf("%d", &f);

    // Ini alize frames to empty (-1)
    for (i = 0; i < f; i++)
frames[i] = -1;

    // Process each page in the reference string
    for (i = 0; i < n; i++) {
flag = 0;

        // Check if page is already in a frame
        for (j = 0; j < f; j++) {
if (frames[j] == pages[i]) {
flag = 1;                break;
            }
        }

        // If page not found (page fault)
        if (flag == 0) {
if (i < f) {
                // Fill empty frame slots first
frames[i] = pages[i];                faults++;

            } else {
                // Find the page to replace (Op mal strategy)
                farthest = -1;
```

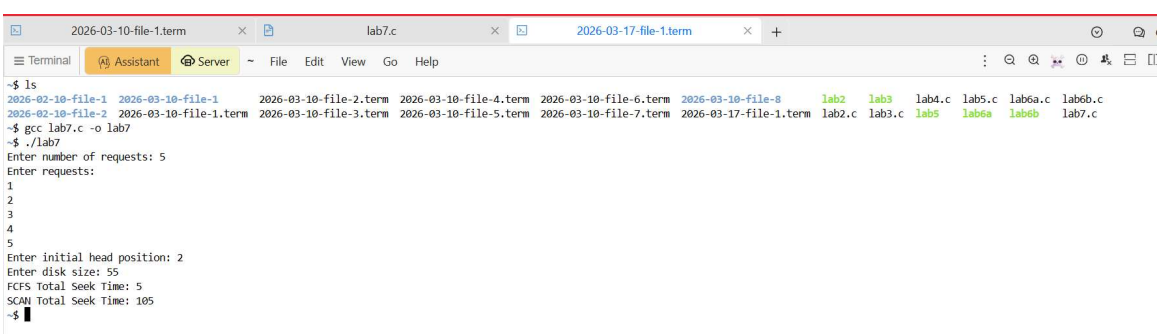
```
            for (j = 0; j < f;
j++) {                  index = -
1;

                // Look ahead in the reference string
for (k = i + 1; k < n; k++) {                   if
(frames[j] == pages[k]) {
index = k;                      break;
                  }
                }

                // If a page is never used again, replace it
if (index == -1) {                              pos = j;
break;
                }

                // Otherwise, replace the one used farthest in future
if (index > farthest) {                     farthest = index;
pos = j;
                }
              }

              frames[pos] = pages[i];
faults++;

          }
       }
   }

   // Output total page faults     prin ("Total
Page Faults = %d\n", faults);
```

```
    return 0;
}
```




LAB7

```
#include <stdio.h>

int main() {    int pages[50], frames[10];
    int n, f, i, j, k, flag, faults = 0;    int pos, farthest, index;

    // Input: number of pages    prin ("Enter number of pages: ");    scanf("%d", &n);
```

```
// Input: page reference string     prin ("Enter page reference string:\n");
for (i = 0; i < n; i++)        scanf("%d", &pages[i]);

// Input: number of frames     prin ("Enter number of frames: ");     scanf("%d", &f);

// Ini alize frames to empty (-1)
for (i = 0; i < f; i++)        frames[i] = -1;

// Process each page in the reference string
for (i = 0; i < n; i++) {        flag = 0;

    // Check if page is already in a frame
    for (j = 0; j < f; j++) {            if (frames[j] == pages[i]) {                flag = 1;                break;
        }
    }

    // If page not found (page fault)
    if (flag == 0) {            if (i < f) {
            // Fill empty frame slots first                frames[i] = pages[i];                faults++;
        } else {
            // Find the page to replace (Op mal strategy)
            farthest = -1;
             for (j = 0; j < f; j++) {                    index = -1;

                // Look ahead in the reference string                    for (k = i + 1; k < n; k++) {
if (frames[j] == pages[k]) {                        index = k;                        break;
                    }
                }
```

```
            // If a page is never used again, replace it                if (index == -1) {
pos = j;                    break;

            }


            // Otherwise, replace the one used farthest in future                if (index > farthest) {
farthest = index;                    pos = j;

            }

          }


          frames[pos] = pages[i];              faults++;

        }

      }

    }


    // Output total page faults     prin ("Total Page Faults = %d\n", faults);


    return 0;

}
```